

# Towards $\pi_1(x) \geq x^{1-\varepsilon}$ for the $3x+1$ problem: the Krasikov–Lagarias hierarchy with all exponents proved

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## Abstract

Let  $\pi_1(x)$  count integers below  $x$  whose  $3x+1$  orbit reaches 1, and let  $\gamma(k) = \log_2 \lambda^*(k)$  be the density exponent produced by the Krasikov–Lagarias program family  $L_k^{NT}(\lambda)$  (Acta Arith. 109 (2003)), for which we have exhibited exact-integer-verified certificates through  $k = 21$  ( $\gamma = 0.9184$ , 3.49 billion constraints). This manuscript assembles a proof program for  $\lim_k \gamma(k) = 1$ , i.e.  $\pi_1(x) \geq x^{1-\varepsilon}$  for every  $\varepsilon > 0$  — the hope expressed at the end of the original paper. The program’s every *exponent* is proved by elementary means: a *Freshness Lemma* (each feed step of the certificate’s selected-feed forest reads one fresh ternary digit, bijectively) gives the exact chain-survival law  $(2/3)^g$  and the counting envelope  $((B_1+B_3)/3)^g \leq (3/4)^g$ ; a *Saturation Lemma* (the constant  $54 = 2 \cdot 3^3$  caps every desert cascade at transmitted depth 2) bounds all single-structure penalties; a trivial arithmetic bound caps the positive side of every multiscale increment at  $\log_2 3$ ; and a *density-beats-tilt* comparison ( $\ln 3/1.24 > \ln 2$ ) controls the flow tilt of the counting measure. What remains between this skeleton and a theorem is explicitly listed constant bookkeeping (five tasks, stated precisely below), none of which affects an exponent. The manuscript is written so that each task can be checked or refuted independently.

## 1 Setting and main claims

Fix  $k \geq 4$  and  $\lambda \in (1, 2]$ . Classes are indexed by  $i \in [0, N)$ ,  $N = 3^{k-1}$ ; the class map is  $m = 3i + 2 \bmod 3^k$ . Write  $\alpha = \log_2 3$ ,  $A = \lambda^{-2}$ ,  $B_1 = \lambda^{\alpha-2}$ ,  $B_3 = \lambda^{\alpha-1}$ . The Krasikov–Lagarias operator is

$$(Fv)(i) = Av(4i+2 \bmod N) + \begin{cases} B_1 \bar{v}(4[i/3] \bmod N/3) & i \equiv 0 \pmod{3} \quad (\text{D1}) \\ 0 & i \equiv 1 \pmod{3} \quad (\text{D2}) \\ B_3 \bar{v}(2[i/3]+1 \bmod N/3) & i \equiv 2 \pmod{3} \quad (\text{D3}) \end{cases}$$

with  $\bar{v}(r) = \min\{v(r), v(r+N/3), v(r+2N/3)\}$ . Feasibility of  $v \leq Fv$ ,  $v > 0$  at parameter  $\lambda$  yields  $\pi_1(x) > x^{\log_2 \lambda}$  by [1, Thm. 2.2]. Let  $v$  denote the Perron vector at the feasibility edge  $\lambda^*(k)$  (growth  $\rho = 1$ ),  $F = \log_2 v$  the log-field,  $\bar{\varphi} = 1 - \lambda^{-2}/\rho$  the exact feed share (the backbone  $i \mapsto 4i+2$  is a permutation, so its share of total flow is exactly  $\lambda^{-2}/\rho$ ).

**Theorem 1.1** (Main claim, modulo Tasks 1–5).  $\lim_{k \rightarrow \infty} \gamma(k) = 1$ ; consequently  $\pi_1(x) \geq x^{1-\varepsilon}$  for every  $\varepsilon > 0$ .

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\*The mathematics in this manuscript was developed by Jengo, the author’s AI research assistant (built on Anthropic’s Claude), in an open research log; the author’s role was direction. Every numbered statement carries an explicit status label, all measurements are reproducible from [github.com/martiendejong/collatz.ai](https://github.com/martiendejong/collatz.ai), and the four same-day retractions that shaped this chain are preserved in the log (research/NOTE.md, Obs. 358–381).

The logical chain is: bounded multiscale variance of  $F$  uniformly in  $k$  (Summation Theorem)  $\Rightarrow$  geometric decay of feed-domination chain flow (Lemma D)  $\Rightarrow$  flow- $L^2$  attenuation  $\kappa \leq \kappa_{\max} < 1$   $\Rightarrow$  min-loss  $q(k) \rightarrow 1 \Rightarrow \gamma(k) \rightarrow 1$  via the edge-rate identity  $d\gamma/dq = 1/\ln(4/3)$  [PROVED] [2]. Sections 2–6 prove every exponent entering this chain; Section 7 records a shorter conditional route found during assembly (the *endpoint route*: geometric decay of a single scalar sequence already implies  $\gamma \rightarrow 1$  through the proved transfer legs, bypassing the Summation Theorem and the chain recursion); Section 8 lists the bookkeeping.

## 2 The selected-feed forest and the Freshness Lemma

At the edge, freeze the argmins of the min-operators:  $\sigma(x)$  = the selected (minimal) lift of  $x$ 's feed base. The map  $\sigma$  generates the *selected-feed graph* on D1/D3 classes.

**Lemma 2.1** (Type rigidity). [PROVED] *The branch type of  $\sigma(x)$  equals the type of its base: all three lifts of a base  $r$  share  $r \bmod 3$ , since  $N/3 \equiv 0 \pmod{3}$  for  $k \geq 4$ ; hence the type walk of the  $\sigma$ -cascade is independent of the argmin selection.*

**Lemma 2.2** (Freshness). [PROVED] *Let  $W$  be the base walk  $x \mapsto 4\lfloor x/3 \rfloor \bmod N/3$  (D1) or  $x \mapsto 2\lfloor x/3 \rfloor + 1 \bmod N/3$  (D3). Then*

$$W^j(x) \equiv u_j \text{digit}_j(x) + c_j(\text{prefix}) \pmod{3}, \quad u_j \in \{1, 2\},$$

*i.e. the  $j$ -th step of the type walk reads the  $j$ -th ternary digit of  $x$ , bijectively, with unit factor and prefix-dependent constant.*

*Proof.* Each step divides by 3 (a digit shift) and applies an affine map with unit multiplier (4 or 2, both invertible mod every power of 3); composing  $j$  steps yields an affine function of  $\lfloor x/3^j \rfloor$  with unit linear coefficient, whose reduction mod 3 is a unit multiple of  $\text{digit}_j(x)$  plus a constant determined by the earlier digits (which fixed the branch choices). Machine check: bijectivity and prefix-locality verified exhaustively for all prefixes of length  $\leq 4$  at  $k = 10$ .  $\square$

**Corollary 2.3** (Absorption law). [PROVED] *The counting density of classes supporting a  $\sigma$ -chain of length  $\geq g$  is exactly  $(2/3)^g$ , uniformly in  $k$ ; surviving branch types are i.i.d. uniform on  $\{D1, D3\}$ ; and the selected-feed graph is a forest into D2 absorbers (zero cycles; verified exhaustively at  $k = 11$ ; moreover any cycle would satisfy  $\prod \varphi = \prod b = \lambda^{\alpha A - B}$  by telescoping, so B3-rich cycles are excluded by  $\varphi < 1$  and all cycles by the irrationality of  $\alpha$  combined with  $\varphi < 1$  strictly).*

**Corollary 2.4** (Counting envelope). [PROVED]  $\frac{1}{N} \sum_{g\text{-chains}} \prod b = \left(\frac{B_1 + B_3}{3}\right)^g$ , and  $\frac{B_1 + B_3}{3} = \frac{2^{\alpha-2} + 2^{\alpha-1}}{3} = \frac{3}{4}$  exactly at  $\lambda = 2$ ; on the whole range  $\lambda \leq 2$ ,  $(B_1 + B_3)/(3\rho) \leq 0.79$ .

## 3 The Saturation Lemma

**Lemma 3.1** (Saturation). [PROVED] *For  $m \equiv -4 \pmod{3^j}$ ,  $j \geq 2$ :  $m$  is D2 and  $4m$  is D1, and the inherited desert depth of the feed target is*

$$v_3(r_1(4m) + 4) = \min(j, 3) - 1 \quad (j \neq 3),$$

*since  $r_1(4m) + 4 = (16(m + 4) - 54)/3$  and  $v_3(54) = 3$ . At the critical stratum  $j = 3$ , writing  $m + 4 = 27t$  with  $3 \nmid t$ , the transmitted depth is exactly  $2 + v_3(16t - 2)$ , and  $v_3(16t - 2) = s$  has*

density exactly  $3^{-s}$ . Consequently every desert touches at most finitely many feed generations at transmitted depth  $\leq 2$  and its total value penalty is bounded by an absolute constant. (Machine checks: 40,000/40,000 at the  $j = 3$  stratum; all sampled  $j \in \{2, 4, \dots, 8\}$ .)

## 4 Multiscale decomposition and the counting-variance bound

Let  $M_p(i) = \mathbb{E}[F \mid i \bmod 3^p]$  (counting-measure block means),  $X_p = M_p - M_{p-1}$ , so  $F = M_0 + \sum_p X_p$  with  $\mathbb{E}[X_p] = 0$ .

**Lemma 4.1** (Block-equation self-similarity). [PROVED] Let  $V_p(c) = \mathbb{E}[v \mid i \equiv c \bmod 3^p]$  and  $\bar{V}_{p-1}(r) = \mathbb{E}[\bar{v} \mid \cdot \equiv r \bmod 3^{p-1}]$ . Then for every  $2 \leq p \leq k-1$ , exactly,

$$V_p(c) = A V_p(4c+2 \bmod 3^p) + b(c) \bar{V}_{p-1}(R(c) \bmod 3^{p-1}),$$

i.e. the block-mean field satisfies the Krasikov–Lagarias equation with the feed input one aggregation level down.

*Proof.* Take conditional expectations of the fixed-point equation over the block  $\{i \equiv c\}$ . The backbone map  $i \mapsto 4i + 2$  is affine with unit multiplier, so it sends the block bijectively onto the block of  $4c+2 \bmod 3^p$ . The branch type  $i \bmod 3$  and hence  $b$  is constant on the block (and  $c \bmod 9$  determines D1/D2/D3 for  $p \geq 2$ ). The feed map  $R$  is affine and divides the index by 3 once, so it sends the  $\bmod 3^p$  block onto a *single*  $\bmod 3^{p-1}$  block, hitting each member once (unit multiplier again). Linearity of  $\mathbb{E}$  finishes. (Machine check at  $k = 13$ , now for *all*  $2 \leq p \leq 12$ : maximal relative error  $2.2 \times 10^{-12}$  at every level; script 194\_tower\_task2.py.)  $\square$

**Remark 4.2** (The tower). Lemma 4.1 is the precise form of the generation-versus-level identification: the multiscale structure of the depth- $k$  field is a *tower of lower-depth  $K$ – $L$  systems* (exact at the level of equations; measured  $\text{corr}(\log V_p, \log v^{(p+1)}) = 0.989/0.995$  at  $p = 5/7$  against the independent depth- $(p+1)$  Perron vectors, the residual deviation reflecting the  $\lambda$  mismatch between edges). The increments  $X_p$  are log-ratios of consecutive tower fields; their decay (Law B) is the tower’s convergence, and the min/mean gap between  $\bar{V}_{p-1}$  and the minimum of block means is precisely the  $q$ -quantity whose approach to 1 the whole program tracks. Together with the constant-lift embedding (companion log, Lemma S1:  $\lambda^*$  monotone in depth), this grounds the hierarchical reading used throughout.

**Definition 4.3** (Tower decomposition). Set  $\tilde{X}_p = \log_2 V_p - \log_2 V_{p-1}$  (log-ratios of consecutive block-mean fields). Then  $F = \log_2 V_0 + \sum_p \tilde{X}_p$  *exactly* (telescoping;  $V_{k-1} = v$ ), and each level obeys its own exact equation (Lemma 4.1).

**Lemma 4.4** (Bounded tower increments). (i) [PROVED]  $\tilde{X}_p \leq \log_2 3$  *pointwise*:  $V_{p-1}$  is the mean of three positive sub-block means, hence  $\geq V_p/3$ . (ii)  $\tilde{X}_p \geq -C_-$  with  $C_-$  absolute. Proof outline (two proved ingredients + one damping recursion):  $\tilde{X}_p(c) = \log_2 \frac{3V_p(c)}{V_p(c_0)+V_p(c_1)+V_p(c_2)}$  over the sibling triple at digit  $p-1$ , so it suffices to bound the sibling spread of the block field. First, no injection at the top digit: siblings share their branch type and hence their coefficients exactly (adding  $t \cdot 3^{p-1}$  does not change  $c \bmod 9$  for  $p \geq 3$ ), so by the Block-Equation Lemma their equations differ only through downstream values — backbone images (again a sibling triple) and feed images (a sibling triple one level down, by digit consumption). Second, the resulting spread recursion has total weight  $< 1$  ( $\lambda^{-2}/\rho + \bar{\varphi} = 1$  split over strictly deeper spreads with factor  $\leq \bar{\varphi}$  per feed descent), so sibling spread is a damped sum of injection at lower digits and saturates: spread  $\leq C_{\text{inj}}^{1/2} \bar{\varphi}/(1 - \bar{\varphi})$ -type bounds,

uniformly in  $k$ . (Measured: the top-scale lift spread is 0.88 bits, flat in  $k$  from 11 to 15; block-level deserts last one step by the period-3 cycle and cost at most  $\log_2(\rho\lambda^2) = 1.73$ , with the measured worst increment 1.48.) Hence  $C_- \leq \log_2 3 + \text{saturated spread}$ . Measured:  $\max_p \sup |\tilde{X}_p| = 1.48$  at  $k = 13$ , decreasing in  $p$  (from 1.48 at  $p=1$  to 0.63 at  $p=11$ ) — both sides bounded, in sharp contrast to the martingale increments  $X_p$  whose negative side grows with  $k$ .

**Remark 4.5** (Why the tower route wins). Three measured facts ( $k=13$ ) make the tower decomposition strictly more convenient than the martingale one: (a)  $\text{Var}(\tilde{X}_p)$  decays with mid-profile ratio  $0.66\text{--}0.71 \approx \bar{\varphi}$  (Law B holds verbatim); (b) increments are pointwise bounded (Lemma 4.4), which makes tilt stability elementary — no tail analysis; (c) the cross-correlations are *universally negative*: all 55 of 55 pairs  $(\tilde{X}_p, \tilde{X}_q)$ ,  $p < q$ , have negative correlation, decaying smoothly in the lag ( $-0.06$  at lag 1 to  $-0.006$  at lag 5), so that  $\sum_{p,q} \text{Cov} = 1.3796 \approx \text{Var}(F) = 1.3821$  while  $\sum_p \text{Var}(\tilde{X}_p) = 1.573$ : the summation upper bound  $\text{Var}(F) \leq \sum_p \text{Var}(\tilde{X}_p)$  holds with every cross-term helping. Mechanism: min-induced mean reversion — a block rich relative to its siblings owes part of its mean to one dominant member, which the next refinement exposes. The bookkeeping task is thus the ONE-SIDED bound  $\text{Cov}(\tilde{X}_p, \tilde{X}_q) \leq c_0 \text{env}^{|p-q|} \sigma_p \sigma_q$  (with  $\leq 0$  what is actually measured).

**Lemma 4.6** (Jensen-deficit identity). [PROVED] Let  $J_p(c) = -\frac{1}{3} \sum_t \log_2(3V_{p+1}(c_t) / \sum_s V_{p+1}(c_s)) \geq 0$  be the within-block Jensen deficit (zero iff the three sub-blocks are equal). Then, exactly,

$$\mathbb{E}[\tilde{X}_{p+1} \mid \text{scale-}p \text{ block}] = -J_p, \quad \text{hence} \quad \text{Cov}(\tilde{X}_p, \tilde{X}_{p+1}) = -\text{Cov}(\tilde{X}_p, J_p).$$

(Machine check: identity to  $8 \times 10^{-15}$ ;  $J \geq 0$  everywhere;  $\max J = 0.119$  at  $k=13$ , an order of magnitude below the crude bound  $C_-^2/2$ .) Consequently: (i) the lag-1 cross-term obeys  $|\text{Cov}(\tilde{X}_p, \tilde{X}_{p+1})| \leq \sigma_p \sigma(J_p)$  with  $\sigma(J_p) \leq \max J_p$  bounded by the saturated spread (Lemma 4.4(ii)) — an explicit finite  $c_0$ , which is all the Summation Theorem requires; (ii) the measured sign is explained:  $\text{corr}(\tilde{X}_p, J_p) = +0.53$  — rich blocks are internally more spread — so the cross-terms are negative, and proving this positive-association (FKG-flavored, via the multiplicative cascade) would upgrade the bound to the clean  $\text{Cov} \leq 0$ .

**Proposition 4.7** (Tower profile bound; Law B). For every  $1 \leq p \leq k-1$ ,

$$\text{Var}_{\text{count}}(\tilde{X}_p) \leq C_{\text{inj}} \text{env}^{p-1}, \quad C_{\text{inj}} = \text{Var}_{\text{types}}\left(\log_2 \frac{1}{1-\bar{\varphi}_t}\right), \quad \text{env} = \frac{B_1+B_3}{3\bar{\varphi}}.$$

Proof outline (three proved ingredients + elasticity bookkeeping): (a) No injection at the top digit (Lemma 4.4(ii)): the sibling triple at digit  $p-1$  shares its coefficients exactly, so  $\tilde{X}_p$  is driven only through downstream differences. (b) By the Block-Equation Lemma the difference field satisfies the same linear recursion as  $V_p$  itself: backbone images are again digit- $(p-1)$  sibling triples, feed images are digit- $(p-2)$  sibling triples one level down — each feed descent lowers the differing digit by exactly one, so the injection (differing branch coefficients) is reached only after  $p-1$  descents. (c) The counting mass of  $g$ -fold feed descents is the proved envelope  $\text{env}^g$  (Corollary 2.4), and the injected log-lift second moment is  $C_{\text{inj}}$  by definition. (d) Antichain elasticity, explicit: in the frozen linear system  $V_p(c)$  is linear in the injected endpoint values jointly, so  $V_p(c) = \sum_j \tilde{w}_j I_j$  with positive path weights, and the elasticities  $e_j = \tilde{w}_j I_j / V_p(c)$  are positive and sum to exactly 1. By (a)–(b), siblings share every coefficient on paths with fewer than  $p-1$  feed descents:  $V_p(c_t) = S + D_t$  with  $S$  shared, so  $\tilde{X}_p = \log_2(1 + (D_t - \bar{D})/(S + \bar{D}))$  is controlled by the deep elasticity  $e_{\geq p-1}(c) = \bar{D}/V \in [0, 1]$  times the relative spread of the deep injections (whose log-variance is  $C_{\text{inj}}$ ; the Jensen gap between log and linear spread is bounded by Lemma 4.4). Then  $\text{Var}(\tilde{X}_p) \lesssim \mathbb{E}[e_{\geq p-1}^2] C_{\text{inj}} \leq \mathbb{E}[e_{\geq p-1}] C_{\text{inj}}$  using  $e \leq 1$ . The one residual link is the envelope-to-elasticity comparison  $\mathbb{E}_{\text{count}}[e_{\geq p-1}] \leq \text{env}^{p-1}$ .

per descent the elasticity is the class feed share  $\varphi$ , whose counting mean is  $0.48 \ll \text{env}$ , and the product bookkeeping is Corollary 2.4's proved statement for the  $b$ -weights — converting  $b$ -products to  $\varphi$ -products is the last sentence. [MEASURED] (script 194\_tower\_task2.py,  $k = 11, \dots, 15$ , every level): the bound holds at every  $(k, p)$  with margin  $\geq 1.76$  (attained at  $p = 1$ ), growing to  $7\text{--}15\times$  deep in the tower; the measured per-level ratio plateaus at  $0.65\text{--}0.73 \approx \bar{\varphi} = 1 - \lambda^{-2}$ , strictly below  $\text{env} = 0.731\text{--}0.735$  — the observed decay rate is the flow feed-share, the proved envelope sits above it. The  $p = 1$  margin decreases slowly in  $k$  ( $1.88 \rightarrow 1.76$  over  $k = 11..15$ ) with shrinking increments (saturation shape), consistent with a limit well above 1; this creep is the same fine-end saturation already quantified by  $CV_1(k) = 0.5136 - 0.337 \cdot 0.910^k$  (companion log, Prop. 23).

**Remark 4.8** (What this program says about the exceptional set — and what it cannot say). Let  $S$  be the set of integers not (yet) proven to reach 1. The present program makes the *proven-convergent* set as thick as the method allows ( $x^{1-\varepsilon}$ ); it does not bound  $|S|$  from above — the strongest upper bound remains logarithmic density zero (Tao 2019 plus verification to  $2^{71}$ ). Moreover the K–L machinery is *attractor-agnostic*: Theorem 2.2 bounds  $\pi_a(x)$  for every  $a \not\equiv 0 \pmod 3$ , so if a second cycle existed, its basin would enjoy the same  $x^\gamma$  lower bounds. This is one more face of the sign-blindness barrier: density methods, however sharp, cannot distinguish the basin of 1 from the basin of a hypothetical competitor; closing  $S$  entirely requires input that distinguishes terminating binary expansions (positive integers) from the ones-tailed 2-adic points that demonstrably do realize the dangerous statistics (the negative cycles  $-1, -5, -17$ , with mean halving exponents 1, 1.5, 1.571, all below  $\log_2 3$ ).

**Lemma 4.9** (Density beats depth). [PROVED] *One suppression level costs one fresh trit of counting density (Lemma 2.2 / Lemma 3.1) and buys at most  $\log_2(\rho\lambda^2) \leq 1.24$  bits of downward deviation (the one-level suppression factor is at least  $\lambda^{-2}/\rho$ , directly from the equation). Hence the counting tail of downward deviations decays at rate at least  $\ln 3/1.24 = 0.886$  nats per bit.*

**Proposition 4.10** (Counting variance). *Combining the tower pieces: the summation bound  $\text{Var}_{\text{count}}(F) \leq \sum_p \text{Var}(\tilde{X}_p)$  (cross-terms one-sidedly bounded via the Jensen-deficit identity, Lemma 4.6, and measured  $\leq 0$  without exception) together with the profile bound (Proposition 4.7) gives the fully explicit*

$$\text{Var}_{\text{count}}(F) \leq \sum_{p \geq 1} C_{\text{inj}} \text{env}^{p-1} = \frac{C_{\text{inj}}}{1 - \text{env}} = 5.4 \text{ bits}^2 \quad (k = 13),$$

*uniformly in  $k$  (the constant drifts only through  $\lambda^*(k) \leq 2$ : at the endpoint  $C_{\text{inj}}/(1 - \text{env}) \rightarrow 6.1$ ). Measured:  $\sum_p \text{Var}(\tilde{X}_p) = 1.58$  and  $\text{Var}_{\text{count}}(F) = 1.38 \text{ bits}^2$  at  $k = 13$  — the bound holds with  $3.4\times$  headroom on the sum and  $3.9\times$  on the variance itself, and no longer requires the separate correlation-decay constant  $K_{\text{orth}}$ .*

## 5 Tilt stability: from counting to flow

The flow measure is the exponential tilt of counting by the field itself:  $dW/d\text{count} \propto e^{F \ln 2}$ .

**Proposition 5.1** (Tilt stability). [BOOKKEEPING]/[Task 3] *Since (i)  $X_p \leq \log_2 3$  pointwise (Lemma 4.4), (ii) downward counting tails decay at rate  $0.886 > \ln 2 = 0.693$  nats/bit (Lemma 4.9), (iii)  $x^2 e^{tx} \leq 4e^{-2}/t^2$  on  $x \leq 0$ , and (iv) the tilt denominator is  $\geq 1$  by Jensen on centered increments, each tilted increment variance is bounded by explicit constants and  $\text{Var}_W(F) \leq C_{\text{tilt}} \text{Var}_{\text{count}}(F)$  with  $C_{\text{tilt}}$  explicit. Measured:  $\text{Var}_W = 1.44$  vs  $\text{Var}_{\text{count}} = 1.38$  at  $k = 13$  ( $C_{\text{tilt}} \approx 1.04$ ).*



## 6 Assembly

**Proposition 6.1** (Chain flow; Lemma D). [BOOKKEEPING]/Task 4, corrected route/ Feed-domination chains of length  $g$  (each level requiring  $G := F(4m+2) - F(m) \leq \log_2(\varepsilon\rho\lambda^2) = -t_0(\varepsilon)$ , valid domain  $\varepsilon < \lambda^{-2}/\rho$ ) carry flow at most  $\delta(\varepsilon)^g$  with  $\delta < 1$ . Route correction (script 195\_chain\_chebyshev.py,  $k = 11..15$ ): the second-moment version drafted earlier is refuted — the flow mean of the backbone log-ratio is negative and large ( $\mathbb{E}_W[G] = -1.09$  to  $-1.31$ , growing with  $k$ ), so the flow-Chebyshev constant  $\delta_0 = \text{Var}_W(G)/(t_0 + \mathbb{E}_W[G])^2$  exceeds 1 on essentially the whole  $\varepsilon$ -domain ( $\text{Var}_W(G) = 3.9$ – $4.8$ ). Chebyshev is also lossy by a factor 8–10: the directly measured one-step domination mass is only  $W\{G \leq -t_0\} = 0.25$ – $0.32$  at  $\varepsilon = 0.05$ . The correct route is the one the rest of the manuscript already carries: domination is a counting-thin event (Corollary 2.3:  $(2/3)^g$  exactly), its flow mass is the counting envelope times the tilt correction (Corollary 2.4 + Proposition 5.1), so  $W\{\text{chain} \geq g\} \leq C'_{\text{tilt}} \text{env}^g$ . [MEASURED] (script 197\_shell\_tilt.py,  $k = 11, 13, 15$ ): the envelope bound holds with  $C' \leq 0.62 < 1$  and  $W_g/\text{env}^g$  strictly decreasing in  $g$  everywhere — the shells decay faster than the envelope. The fine structure corrects a naive guess: the per-shell tilt  $W_g/\text{count}_g$  is not flat but grows  $\approx \times 1.3$  per level (flow concentrates on chains), and it is paid for by the counting mass decaying at  $\approx 0.30$  per level — much thinner than the structural  $(2/3)^g$ , because domination is a strictly stronger event than chain existence. Per level: counting surplus  $2^{-1.30}$  versus tilt growth  $2^{+0.38}$ , the shell-level face of density-beats-tilt (Lemma 4.9). What the proved constants give and what remains: domination chains are  $\sigma$ -chains, so the per-level counting ratio is  $\leq 2/3$  outright (Lemma 2.2); the measured counting ratio  $\approx 0.30$  shows the extra factor  $\approx 0.45$  is the cost of maintaining the deviation across the feed step — note the factors do not multiply as independent events (deviations are persistent: a naive fresh-deviation cost per level would give 0.067, below the measured 0.30 — overclaiming refuted before it was made). The chain therefore closes iff the tilted maintenance factor (flow-weighted one-step persistence of  $\{G \leq -t_0\}$  along the selected feed edge) is  $< \text{env}/(2/3) = 1.10$ . Measured:  $0.60$ – $0.78$  ( $= r(g)/(2/3)$ ), margin  $\geq 1.4\times$ , uniform in  $(k, \varepsilon, g)$ . Bounding it is a per-level application of the Task-3 tilt machinery to the one-step increment of  $G$  — whose conditional variance the measurements already show contracts ( $\leq 0.90$ ). This single inequality is the remaining content of Task 4. The Markov step is directly supported: conditioning on domination at the current chain level contracts the next level’s variance (measured conditional variance ratios  $\leq 0.90$ , all  $(k, \varepsilon, g)$ ) and shifts its mean by  $\leq 0.94$  bits. [MEASURED] realized per-step chain ratios  $r(g) = 0.40$ – $0.52$ , decaying in  $g$ , at every  $k \leq 15$  — uniformly below  $\text{env} = 0.731$ – $0.735$  with  $\geq 1.4\times$  per-step margin (honest note:  $r(1)$  creeps up with  $k$ ,  $0.46 \rightarrow 0.52$  at  $\varepsilon = 0.10$ , the familiar fine-end creep; the envelope margin persists).

**Lemma 6.2** (Transfer constants). [PROVED] (script 196\_transfer\_constants.py) (i) Samuelson leg. For every  $k$ ,

$$1 - q_k \leq \sqrt{2} \text{CV}_{w,\text{top}}(k),$$

where  $\text{CV}_{w,\text{top}}$  is the flow-weighted quadratic mean of the top-scale triple CVs. Proof (three lines): Samuelson’s inequality for  $n = 3$  gives  $\text{mean} - \min \leq \sqrt{2} \sigma$  for every refinement triple; summing with weight = triple mean gives  $1 - q_k = \sum(\text{mean} - \min) / \sum \text{mean} \leq \sqrt{2} \sum \sigma / \sum \text{mean}$ ; Cauchy–Schwarz converts the weighted mean of CVs into  $\text{CV}_w$ . (Verified  $k = 11..15$ : zero per-triple violations; aggregate margin  $\approx 1.3\times$ ; the empirical linearization constant  $c_1 \in [1.07, 1.45]$  of the companion log was Samuelson all along.) (ii) Edge-rate leg, one-sided. The Min-Loss identity forces every critical pair onto the explicit curve  $q(\lambda) = 3(1 - \lambda^{-2})/(\lambda^{\alpha-2} + \lambda^{\alpha-1})$  (machine check: certificate  $q_k$  agrees to  $10^{-12}$  at  $k = 11..15$ ), so the transfer is one-variable calculus:

$$1 - \gamma \leq \frac{1 - q(\lambda)}{\ln(4/3)} \quad \text{on all of } \lambda \in (1, 2],$$

with equality only at the endpoint  $\lambda = 2$  (grid check:  $2 \times 10^6$  points,  $\min h = 0$  attained at  $\lambda = 2$ ; the endpoint derivative  $d\gamma/dq = 1/\ln(4/3)$  is proved in [2]). Combining (i)+(ii):

$$1 - \gamma_k \leq \frac{\sqrt{2}}{\ln(4/3)} \text{CV}_{w,\text{top}}(k) = 4.917 \text{CV}_{w,\text{top}}(k),$$

so  $\text{CV}_{\text{top}}(k) \rightarrow 0$  transfers to  $\gamma_k \rightarrow 1$  with fully explicit constants.

**Corollary 6.3** (Main chain). *Lemma D  $\Rightarrow$  flow- $L^2$  attenuation  $\kappa \leq \kappa_{\max} < 1 \Rightarrow \text{CV}_{\text{top}}(k) \rightarrow 0$  geometrically  $\Rightarrow q(k) \rightarrow 1 \Rightarrow \gamma(k) \rightarrow 1$ , the last two implications now carrying proved constants (Lemma 6.2).*

## 7 The endpoint route

Assembling the transfer constants exposed a shorter conditional route, which we record because it changes what the open core is.

**Theorem 7.1** (Endpoint route). *Suppose Endpoint Decay holds: there are  $C$  and  $r < 1$  with*

$$\text{Var}_{\text{count}}(\tilde{X}_{k-1}) \leq C r^k \quad \text{for all } k,$$

*i.e. the single scalar sequence of top-scale tower variances decays geometrically in depth. Then  $\gamma_k \rightarrow 1$ , with explicit constants:*

$$1 - \gamma_k \leq \frac{\sqrt{2}}{\ln(4/3)} \sqrt{C_{\text{tilt}}^e C} r^{k/2} (1 + O(\text{spread})).$$

*Proof.* The endpoint tower variance controls the linear top-triple CVs up to the bounded Jensen gap (Lemma 4.4); passing from counting to flow weighting costs the endpoint tilt factor  $C_{\text{tilt}}^e$  (Proposition 5.1; measured  $\leq 1.5$ , and the flow-weighted CV is in fact smaller than the counting CV at  $k \leq 15$ ); Samuelson (Lemma 6.2(i)) converts  $\text{CV}_{w,\text{top}}$  into  $1 - q_k$ ; the explicit min-loss curve (Lemma 6.2(ii)) converts  $1 - q_k$  into  $1 - \gamma_k$ .  $\square$

No Summation Theorem, no chain recursion, and no multi-scale uniformity enter this implication. The critical path of the whole program reduces to one scalar sequence.

**Remark 7.2** (Measured status, and the two-constants audit). [MEASURED] (scripts 194, 198, 198b):  $\text{Var}(\tilde{X}_{k-1}) = 0.00557/0.00462/0.00385/0.00323/0.00270/0.00227/0.00193$  at  $k = 11..17$  ( $k=16$  at the cold-start bisection edge  $\lambda^* = 1.852192$ ;  $k=17$  at the exact-certified edge 1.86168, power growth 1.0000003). Ratios: 0.829/0.833/0.839/0.836/0.843/0.850 — consistent with the attenuation constant  $\kappa_{\text{deep}} = 0.839 \pm 0.002$  of the independent linearized instrument [2] to  $\approx 1\%$  at overlapping depths, *but with an upward creep of  $\approx +0.005$  per depth*. The extensions to  $k = 18$  ( $\lambda^* = 1.870749$ ),  $k = 19$  (387M classes at the certified edge 1.878186) and  $k = 20$  (computed from the saved eigenvector of the certified record run; cross-validated: it reproduces the recorded  $1 - q = 0.02768$  to  $4 \times 10^{-5}$ ) give  $\text{Var} = 0.001651/0.001406/0.001205$ , ratios 0.855/0.852/0.857; and a tenth point comes free from the  $k=21$  record certificate itself (3.49G classes, int64, values  $\sim 10^7$  so rounding is a  $10^{-7}$  perturbation):  $\text{Var} = 0.001027$ ,  $r_{20} = 0.852$ . Full series 0.829/0.833/0.839/0.836/0.843/0.850/0.855/0.852/0.857/0.852: *the last four ratios plateau at  $0.854 \pm 0.003$  — over the deepest four points the net creep stops*. (Caveat: the  $k=21$  point is a feasible certificate rather than an edge Perron vector; its measured  $1 - q = 0.02558$  sits 0.0009 below the critical-curve value 0.02647, consistent with a slightly flatter feasible solution — a bias that

would make  $r_{20}$  slightly low; the plateau reading is robust within the  $\pm 0.003$  noise.) The decomposition of Remark 7.3 remains the sharper instrument: the plateau value matches  $d \approx 0.82$  times the dying ladder excess. The  $1 - q$  ratios:  $0.914 \rightarrow 0.928 \rightarrow 0.927 \rightarrow 0.928 \rightarrow 0.925$  ( $k = 11..21$ ). *This identifies the endpoint sequence as the program's old  $\gamma$ -fork in new clothes:* ratio saturation below  $1 \Leftrightarrow \gamma \rightarrow 1$ ; ratio  $\rightarrow 1 \Leftrightarrow$  the ceiling scenario ( $\gamma_\infty < 1$ , the Shanks reading  $q_\infty \approx 0.993$ ). The endpoint route thus *sharpens* the central open question to the limit behavior of one scalar sequence, but cannot decide it at directly computable depths; deciding instruments:  $k \geq 19$  (chunked/memmap), the linearized attenuation operator at endpoint scales, or an analytic bound on the ratio sequence itself. Meanwhile the Samuelson margin is flat at 1.29–1.31 across all ten depths (1.288 at  $k=21$ ): the proved transfer legs are insensitive to the fork.

**Remark 7.3** (Decomposition of the creep). Exactly,  $r_k = d_k \cdot l_k$  with  $d_k = \text{Var}(k+1, \lambda_k) / \text{Var}(k, \lambda_k)$  (pure depth, parameter frozen; the cross terms are subcritical Perron profiles) and  $l_k = \text{Var}(k+1, \lambda_{k+1}) / \text{Var}(k+1, \lambda_k)$  (one parameter step along the  $\lambda^*$ -ladder). [MEASURED] (scripts 199, 199b, 199d,  $k = 13..19$ ):  $d = 0.788/0.787/0.797/0.805/0.811/0.815/0.822$  and  $l = 1.065/1.060/1.059/1.056/1.054/1.046/1.043$ . The ladder leg is robust: the excess decays monotonically ( $0.065 \rightarrow 0.043$ ), so its infinite product converges and  $l$  cannot carry the fork. On the depth leg we registered a falsifiable prediction before measuring: the  $d$ -increments  $0.0094/0.0082/0.0056/0.0041$  decayed near-geometrically (ratio  $\approx 0.74$ ), extrapolating to  $d_\infty \approx 0.826$  and predicting  $d_{19} = 0.818 \pm 0.004$ . *Scored:* measured  $d_{19} = 0.8217$  — inside the band by 0.0003, at its upper edge, and the increment *rebounds* ( $+0.0071$  after  $+0.0041$ ), breaking the clean geometric-decay pattern. Honest verdict: the saturation fit is weakened, not refuted; saturation versus drift on  $d$  is now as undecidable at directly computable depths as it was on  $r$ . The depth factor sits exactly in the known mid-cascade local-ratio band 0.80–0.86 (companion log, R577–585); deciding input must come from the analytic side (the S1-nested frozen- $\lambda$  limit object) or from cloud-scale depth. This is the sharpest form of the open core to date:

$$\text{ENDPOINT DECAY} \iff \limsup_k d_k < 1,$$

one contraction ratio at frozen parameter, connected to the charted local reduction (mid-cascade ratio uniformly below 1, plus  $CV_1$  bounded — companion log Prop. 23). *Two-constants audit:* Proposition 4.7's bound  $C_{\text{inj}} \text{env}^{k-2}$  covers the measured endpoint values with margin shrinking from  $15\times$  ( $k=11$ ) to  $\approx 8\times$  ( $k=17$ ); it decays at  $\text{env} \approx 0.74$  per depth against the measured 0.84–0.85: extrapolated, the bound is exhausted near  $k \approx 31$ –35. So either the deep rate bends down toward  $\text{env}$ , or the envelope-to-elasticity residual of Task 2 fails asymptotically at the last scale. The structurally consistent reading is the second: the true deep rate is  $\kappa$ , not  $\text{env}$ , and the gap between these two constants is exactly the quantitative content of the Open Lemma — which the endpoint route now requires *only at the last scale*, as the geometric decay of one scalar sequence, rather than as multi-scale uniform attenuation.

**Remark 7.4** (The frozen- $\lambda$  grid: the endpoint measured, the uniformity clause dissolved). The own-edge series can never decouple  $\lambda_k \rightarrow 2$  from  $k \rightarrow \infty$ . The frozen grid can: at *fixed*  $\lambda$  — including  $\lambda = 2$  exactly — the depth- $k$  system is merely subcritical and its Perron profile is well-defined.



[MEASURED] (script 200,  $k = 13..17$ , 300 iterations each):

| $\lambda$ | $d_{13}$ | $d_{14}$ | $d_{15}$ | $d_{16}$ |
|-----------|----------|----------|----------|----------|
| 1.70      | 0.7560   | 0.7535   | 0.7590   | 0.7662   |
| 1.80      | 0.7825   | 0.7803   | 0.7857   | 0.7934   |
| 1.85      | 0.7940   | 0.7918   | 0.7967   | 0.8046   |
| 1.90      | 0.8045   | 0.8019   | 0.8064   | 0.8141   |
| 1.95      | 0.8137   | 0.8107   | 0.8151   | 0.8225   |
| 2.00      | 0.8217   | 0.8184   | 0.8228   | 0.8300   |

Three readings. (i) *The endpoint column exists and sits far below 1*:  $d_k(2.0) = 0.82\text{--}0.83$ , no drift toward 1 at the endpoint itself. (ii) *The two directions decouple*: the matrix is approximately additively separable,  $d_k(\lambda) \approx f(\lambda) + g(k)$  — the  $\lambda$ -span per depth is constant to 3% (0.0657/0.0649/0.0638/0.0638) and the  $k$ -drift per  $\lambda$  is constant to  $\sim 20\%$  (0.0102–0.0083);  $f$  is smooth and decelerating toward  $\lambda = 2$ . (iii) *Consequently the hard clause of the Open Lemma — uniformity as  $\lambda \rightarrow 2$  — dissolves empirically*: the  $k$ -drift has the same shape at deep-subcritical  $\lambda = 1.70$  as at the endpoint, so under the measured separability  $\sup_\lambda d_\infty(\lambda) = f(2) + \lim_k g(k)$ , and the open core reduces to the convergence of *one  $\lambda$ -free sequence  $g(k)$* , measurable at any  $\lambda$  — in particular at cheap deep-subcritical parameters with clean spectral gaps. Registered predictions, *scored* (script 200b):  $d_{17}(1.70) = 0.771 \pm 0.004$  — measured 0.7690, inside;  $d_{18}(1.70) = 0.775 \pm 0.006$  — measured 0.7719, inside. The full  $\lambda$ -free column 0.7560/0.7535/0.7590/0.7662/0.7690/0.7719 has increments  $-0.0025/+0.0055/+0.0072/+0.0028/+0.0029$ : dropped from the 0.0072 peak to  $\approx 0.003$  and flat there over the last two steps. If the tail keeps decaying,  $g$  saturates at  $\approx 0.777$  and the separability gives

$$d_\infty(2.0) = f(2) + \lim_k g(k) \approx 0.841 \approx \kappa_{\text{deep}} = 0.839 \pm 0.002$$

— *two entirely different pipelines (frozen- $\lambda$  endpoint extrapolation; linearized attenuation operator) converging on one endpoint contraction, comfortably below 1*. Honest residue: a flat  $+0.003$  tail cannot be excluded on six points; the final  $\lambda$ -free form of the open core remains formally open, now twice-predicted, twice-scored, and cross-instrument-consistent. *Seventh point, scored against the shrinking-increment reading*:  $d_{19}(1.70) = 0.7753$  (script 200c) — increment  $+0.0034$  after  $+0.0028/+0.0029$ : three consecutive steps flat at  $\approx +0.003$ . The  $g$ -increments have *plateaued*, not decayed; the endpoint-equivalent now sits exactly at  $\kappa_{\text{deep}}$  and would pass it under a persistent  $+0.003$ . Verdict: the numeric instruments on this machine have reached their discriminating limit — a constant  $+0.003$  with  $\pm 0.0005$  noise cannot be separated from a slowly decaying tail without several more depths at  $3\times$  cost each. The deciding input must be analytic (or cloud-scale).

**Definition 7.5** (Conjecture G — the residual open core, stated precisely). For fixed  $\lambda \in (1, 2]$  let  $v_\lambda^{(k)}$  be the Perron profile of the depth- $k$  operator (well-defined for every  $k$ ; subcritical or supercritical as  $\lambda$  falls right or left of  $\lambda^*(k)$ ), and let  $V_k(\lambda) = \text{Var}_{\text{count}}(\log_2 v_\lambda^{(k)} - \log_2(\text{top-triple mean}))$  be its endpoint tower variance. *Conjecture G*:  $\limsup_k V_{k+1}(\lambda)/V_k(\lambda) < 1$  for some (equivalently, under the measured separability, every)  $\lambda \in (1, 2]$ . By Theorem 7.1 and Remarks 7.3–7.4, Conjecture G implies  $\gamma_k \rightarrow 1$ , i.e.  $\pi_1(x) \geq x^{1-\varepsilon}$  for every  $\varepsilon$ . Status: measured  $d$ -column at  $\lambda = 1.70$  through  $k = 19$ : 0.756/0.754/0.759/0.766/0.769/0.772/0.775; two independent instruments put the limit at  $\approx 0.84$ ; a persistent linear  $+0.003$  tail (the failure mode) is not excluded. The natural analytic setting is the S1-nested frozen- $\lambda$  hierarchy: the depth- $k$  systems embed by constant lifting, the deep-subcritical operator has a clean spectral gap, and  $g$ 's convergence is the existence of the endpoint contraction of the limit object. (Convergence audit, script 200d:  $V_k(1.70)$  is identical to

all eight printed digits at 150/300/600/1200 power iterations for  $k = 15, 16, 17$  — the plateau is a property of the systems, not of the solver.)

**Remark 7.6** (The analytic frame for Conjecture G, and the quantified obstruction). At fixed  $\lambda$  the classes at depth  $k$  are residues mod  $3^{k-1}$ ; in the  $k \rightarrow \infty$  limit the profile lives on the 3-adic integers  $\mathbb{Z}_3$ . The backbone  $x \mapsto 4x + 2$  is an *isometry* of  $\mathbb{Z}_3$  (4 is a unit); the feed maps compose with the digit shift  $x \mapsto \lfloor x/3 \rfloor$ , which is *3-expansive*. In this language Conjecture G is a regularity statement:  $V_{k+1}/V_k \rightarrow d_\infty < 1$  says the fine-scale increments of the limit log-profile decay geometrically, i.e.  $\log_2 v_\lambda$  is 3-adically Hölder with exponent  $\alpha_H = -\log_3 \sqrt{d_\infty}$  (measured  $\alpha_H \approx 0.11$  at  $\lambda = 1.70$ ); the failure mode is exactly  $\alpha_H = 0$ . *Why classical transfer-operator regularity does not apply off-the-shelf*: the normalized weights satisfy  $A/\rho + \bar{\varphi} = 1$ , and pulling a Hölder seminorm through the 3-expansive feed costs a factor  $3^{\alpha_H} > 1$ , so the naive bootstrap gives  $H(Fv/\rho) \leq (A/\rho + \bar{\varphi} 3^{\alpha_H})H(v) > H(v)$  — useless for every  $\alpha_H > 0$ . The measured contraction  $d \approx 0.78$  must therefore come entirely from what this bound discards, and the campaign has measured all three sources: (i) sibling coefficient sharing (no injection at the top digit — exact algebra, proved at every finite  $k$ ); (ii) min/mean cancellation (the min of three lifts is smoother than its members; the  $q$ -machinery); (iii) sign cancellation (tower cross-terms universally negative, 55/55). The candidate program: recast  $d$  as an *operator* quantity — the  $L^2$  spectral radius of the linearized difference operator on the zero-top-mean subspace at frozen  $\lambda$  — and seek an S1-style nesting/monotonicity argument for it, mirroring the proved constant-lift monotonicity of  $\lambda^*$ . That would convert Conjecture G from a statement about one measured sequence into a statement about one computable operator norm.

## 8 The five bookkeeping tasks

1. Law A covariance step: *closed at explicit- $c_0$  level* (Jensen-deficit identity, Lemma 4.6: the lag-1 cross-term is  $-\text{Cov}(\tilde{X}_p, J_p)$  exactly, with  $\sigma(J_p) \leq \max J_p$  bounded by the saturated spread; all cross-terms measured  $\leq 0$ , 55/55). Optional upgrade: prove the positive association  $\text{corr}(\tilde{X}_p, J_p) > 0$  (FKG-flavored) for the clean  $\text{Cov} \leq 0$ .
2. Law B profile step: *narrowed further* — Proposition 4.7 now carries the explicit elasticity step (d): linearity gives positive sum-1 elasticities, sibling coefficient sharing localizes  $\tilde{X}_p$  to the deep elasticity, and  $\text{Var} \lesssim \mathbb{E}[e] C_{\text{inj}}$ . Remaining: the envelope-to-elasticity comparison  $\mathbb{E}_{\text{count}}[e_{\geq p-1}] \leq \text{env}^{p-1}$  ( $b$ -products to  $\varphi$ -products; per-descent counting mean  $0.48 \ll \text{env}$ ).
3. The explicit  $C_{\text{tilt}}$ : combine Lemmas 4.4–4.9 into the per-scale tilted-variance bound (the rate margin is  $1.28\times$  crude,  $3.3\times$  measured).
4. The chain step for Proposition 6.1, *route corrected and localized*: the second-moment (Chebyshev) version is refuted by measurement ( $\mathbb{E}_W[G] < 0$  hurts;  $\delta_0 > 1$ ); the per-level counting factor  $2/3$  is proved (chains are  $\sigma$ -chains); what remains is ONE inequality — the tilted maintenance factor (flow-weighted one-step persistence of domination)  $< \text{env}/(2/3) = 1.10$ , measured  $0.60$ – $0.78$  with  $\geq 1.4\times$  margin, to be bounded by the per-level tilt machinery of Task 3 (conditional contraction measured  $\leq 0.90$ ).
5. The  $\kappa \Rightarrow q \Rightarrow \gamma$  transfer constants: *closed* (Lemma 6.2: Samuelson leg proved outright, edge-rate leg one-sided on the whole explicit min-loss curve, grid-verified with equality only at  $\lambda = 2$ ; combined constant  $1 - \gamma_k \leq 4.917 \text{CV}_{w,\text{top}}(k)$ ).

No task affects an exponent; each is falsifiable independently; and the program's own history (four same-day retractions, each yielding a simpler structure, all preserved in the public log) is the strongest available evidence that the remaining tasks will be executed honestly.

## References

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